



NOCTIVIS: Drowsiness Detection System

EEG-Based Solution for Driving Safety

Executive Summary

This research proposal outlines the development of **NOCTIVIS**, an innovative EEG-based drowsiness detection system for driving safety. NOCTIVIS aims to provide real-time brainwave analysis to detect driver drowsiness and fatigue, thereby mitigating the risk of accidents caused by impaired driving conditions. This system leverages advancements in brain-computer interface technology to offer a proactive solution for enhancing road safety.

Problem Statement

Driving fatigue is a significant contributor to road accidents globally. Studies indicate that drowsiness and fatigue account for approximately 90% of fatal road accidents worldwide. The consequences of these accidents include loss of life, severe injuries, and substantial economic costs. Developing effective and reliable drowsiness detection systems is crucial for addressing this critical issue and improving road safety for all drivers.

Literature Review Summary

A comprehensive review of 30 peer-reviewed papers published between 2020 and 2025 reveals that EEG alpha, theta, and beta bands serve as reliable markers for drowsiness and fatigue. These studies demonstrate that machine learning techniques, particularly Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, can achieve detection accuracies ranging from 90% to 99% in identifying varying levels of drowsiness based on EEG data. The review highlights the potential of advanced signal processing and machine learning algorithms for developing accurate and real-time drowsiness detection systems.

Proposed Solution

The proposed solution is a web-based dashboard that provides a real-time assessment of driver fatigue levels using a 4-channel EEG system. The system classifies fatigue into five distinct levels:

- **Awake**
- **Mild**
- **Moderate**
- **Severe**
- **Critical**

The dashboard features real-time visualization of brainwave data and provides alerts when the driver's fatigue level reaches a critical threshold. This proactive approach enables timely intervention and prevents potential accidents caused by drowsy driving.

System Architecture

The NOCTIVIS system architecture comprises the following layers:

1. **Data Acquisition:** Acquiring EEG signals using a 4-channel EEG headset.
2. **Signal Processing:** Preprocessing and feature extraction of EEG data to isolate relevant brainwave patterns.
3. **Classification:** Employing machine learning algorithms (CNN/LSTM) to classify fatigue levels based on extracted features.
4. **Application:** Presenting real-time fatigue assessments and alerts through a web-based dashboard.

Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Firebase
- **Visualization:** Chart.js
- **Machine Learning:** Python, TensorFlow/Keras

Research Methodology

The research methodology employs the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) framework to ensure a rigorous and transparent approach to the literature review and data synthesis. This framework guides the systematic identification, selection, and analysis of relevant studies, ensuring the validity and reliability of the research findings.

Expected Outcomes

The expected outcomes of this research project include:

- A fully functional prototype of the NOCTIVIS drowsiness detection system.
- Validated accuracy and reliability of the system in detecting driver drowsiness.
- Publication of research findings in peer-reviewed journals and conferences.
- Potential commercialization and adoption of the system to enhance driving safety.

Next Steps

The next steps involve securing funding, assembling the research team, procuring necessary equipment, and initiating the development and testing phases of the project. Regular progress reports and stakeholder engagement will ensure the successful completion of the research and the realization of its potential impact on road safety.